

Smartphone Usage and Addiction Analysis

Every step, decision, and technical term explained

This PDF explains the final notebook in plain language: what was done, why it was done, how the decisions are backed by data, and how to interpret the final decision tree model.

Item	Value
Dataset	Smartphone_Usage_And_Addiction_Analysis_7500_Rows.csv
Rows and columns	7,500 rows and 16 columns
Prediction target	addicted_label
Final model	Tuned decision tree without gender, addiction_level, or aggregate/ratio engineered features
Final F1-score	95.48%
Final balanced accuracy	91.48%

1. Analysis Objective

The project predicts whether a record belongs to `addicted_label = 1` or `addicted_label = 0` using non-leaking behavioral features. The analysis is predictive, not clinical: it does not diagnose real addiction and does not prove causation.

Target class	Count	Share
<code>addicted_label = 0</code>	2,192	29.23%
<code>addicted_label = 1</code>	5,308	70.77%

Main decisions:

- Use `addicted_label` as the target.
- Use behavioral usage variables as predictors.
- Exclude identifiers, demographic variables with weak value, and target-proxy variables.
- Use class-aware metrics because the target is imbalanced.

2. Data Loading and Data Quality

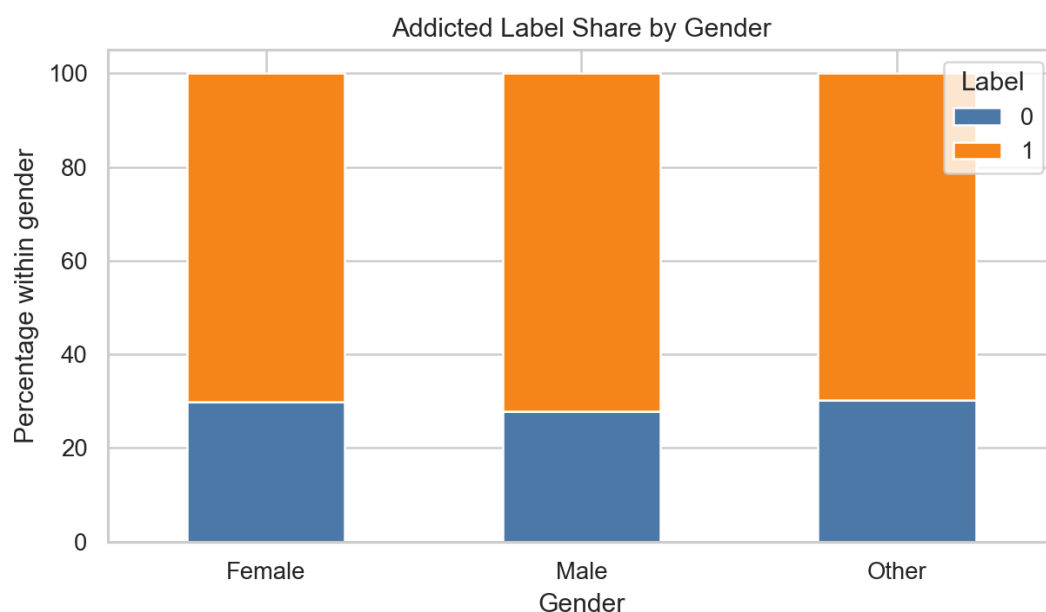
The CSV is loaded into `df_raw`, then a working copy is used for formatting checks. This preserves the original loaded data while allowing standardized text values for analysis.

Check	Result	Decision
Duplicate rows	0	No rows removed.
Duplicate <code>transaction_id</code>	0	Identifier is clean but excluded from modeling.
Duplicate <code>user_id</code>	0	Identifier is clean but excluded from modeling.
Missing values	819 missing <code>addiction_level</code> values	Keep rows; show as Not recorded for EDA only.
Numeric IQR outliers	0 outside IQR bounds for checked numeric fields	No outlier removal needed.

3. Gender Decision

Gender is tested before modeling. The target rates are close across Female, Male, and Other, and Cramer's V is only 0.0232. That indicates a very weak association, so gender is excluded from modeling while all rows are retained.

Gender	Label 0	Label 1	Label 1 share
Female	733	1,728	70.22%
Male	709	1,844	72.23%
Other	750	1,736	69.83%

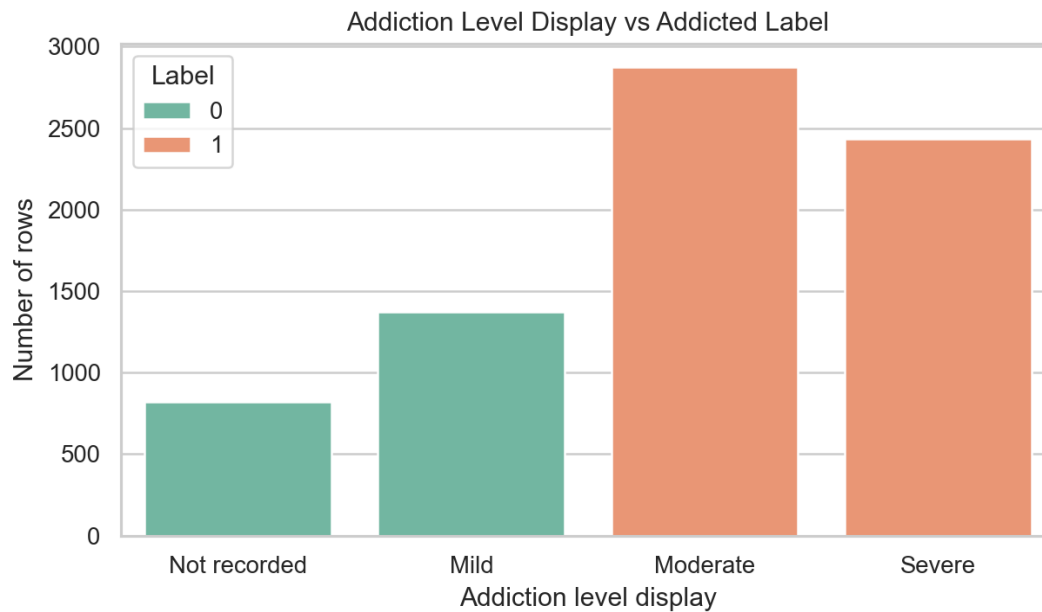


4. Addiction Level Decision

addiction_level is not used as a model input because it behaves like a target proxy. In the data, Not recorded and Mild always map to label 0, while Moderate and Severe always map to label 1.

addiction_level_display	Label 0	Label 1	Interpretation
Not recorded	819	0	Always label 0
Mild	1,373	0	Always label 0
Moderate	0	2,874	Always label 1
Severe	0	2,434	Always label 1

Decision made: keep the rows, display missing addiction_level as Not recorded for EDA only, and exclude addiction_level plus addiction_level_display from modeling.



5. Preprocessing and Feature Preparation

Decision	Reason
Exclude transaction_id and user_id	Identifiers do not represent behavior and can cause memorization.
Exclude addicted_label from X	It is the target, not a predictor.
Exclude addiction_level and display version	They are deterministic target proxies.
Exclude gender	Association with target is weak, and excluding it avoids demographic modeling.
Keep stress and academic/work impact	They are non-leaking categorical predictors.
Create engineered features for exploration	They summarize behavior but are checked in a sensitivity test.

6. Exploratory Data Analysis

Finding	Evidence	Decision impact
Target is imbalanced	70.77% label 1 and 29.23% label 0	Use stratified split and class-aware metrics.
Screen-time variables separate target best	Daily screen time, weekend screen time, and social media hours are higher for label 1	These variables are expected to help the model.
Other variables are weaker	Sleep, notifications, app opens, stress, impact, and gender show smaller differences	Keep some as candidates, but interpret cautiously.
Numeric ranges look plausible	IQR check found no flagged numeric outliers	No row removal needed.

7. Modeling Workflow

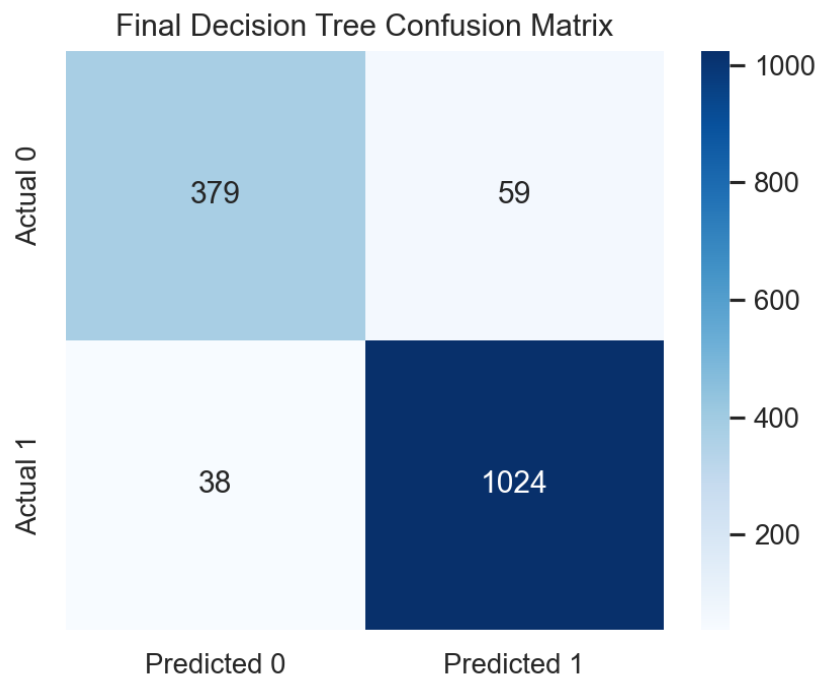
The data is split into 80% training and 20% testing using stratification. Model family selection is performed with 5-fold cross-validation inside the training data, then the selected model is evaluated on the held-out test set.

Model	CV F1	CV macro-F1	CV balanced accuracy
Majority class baseline	0.8288	0.4144	0.5000
Logistic regression	0.9292	0.8755	0.8697
Decision tree	0.9535	0.9208	0.9218
Naive Bayes	0.9042	0.8411	0.8476
KNN	0.9165	0.8490	0.8383
Neural network	0.9366	0.8888	0.8838

8. Recommended Final Model

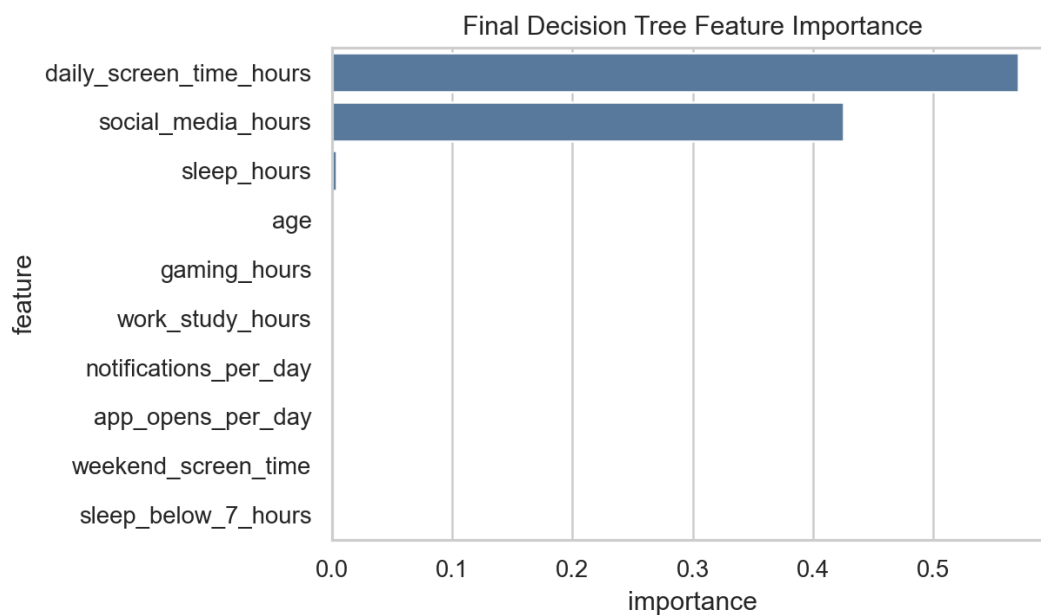
A decision tree won the training-only comparison. A sensitivity check showed that the simpler tree without aggregate and ratio engineered features performed slightly better and is easier to explain, so it is recommended as the final model.

Metric	Value	Meaning
Accuracy	93.53%	Overall share of correct predictions.
Precision for label 1	94.55%	Of predicted label 1 rows, how many were truly label 1.
Recall for label 1	96.42%	Of actual label 1 rows, how many were found.
Recall for label 0	86.53%	Of actual label 0 rows, how many were found.
Balanced accuracy	91.48%	Average recall across both classes.
F1-score	95.48%	Balance of precision and recall for label 1.
Macro-F1	92.07%	Average F1 across both classes.



9. Model Explanation Visuals

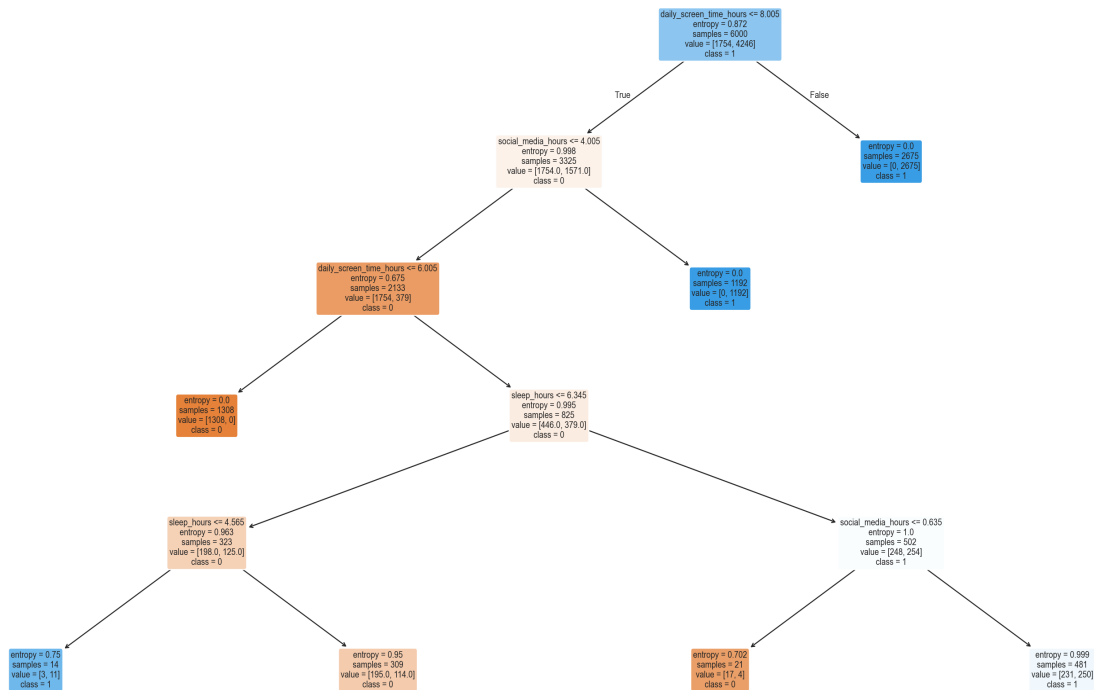
The final tree mostly uses daily screen time and social media hours. Feature importance is model-specific, so it should be read as an explanation of this tree, not as a causal ranking.



9. Model Explanation Visuals - Decision Tree Nodes

A tree node shows the split condition, impurity, number of samples, class counts, and predicted class. The colors show which class a node leans toward.

Recommended Final Decision Tree Node Visualization



10. Decision Log

Decision	Why it was made	Effect on analysis
Keep all gender categories	Other is observed, not missing. Gender showed weak target association.	Preserves all rows and avoids gender imputation.
Exclude gender from modeling	Cramer's V was only 0.0232.	Keeps the model focused on behavior instead of demographics.
Display missing addiction_level as Not recorded	Avoids pretending missing values are a real addiction severity category.	Allows clear descriptive EDA without changing modeling inputs.
Exclude addiction_level	It maps directly to addicted_label.	Prevents leakage and inflated performance.
Use training-only CV	Avoids using the test set to choose the model.	Makes final test performance more trustworthy.
Use simpler final tree	Sensitivity check performed slightly better without aggregate and ratio engineered features.	Improves explainability and defensibility.

11. Limitations

- The dataset is observational, so the analysis cannot prove causation.
- The model should not be used as a medical or clinical diagnosis tool.
- The source and real-world representativeness of the dataset are not independently verified.
- weekend_screen_time has uncertain interpretation and should be described carefully.
- Decision-tree feature importance is model-specific and should not be treated as universal causal importance.

12. Technical Term Glossary

Term	Meaning in this analysis
addicted_label	The binary target being predicted.
addiction_level	A severity label excluded because it behaves like a target proxy.
Baseline model	A simple comparison model, here always predicting the majority class.
Balanced accuracy	Average recall across both classes.
Categorical variable	A variable made of labels or groups.
Chi-square test	A test for whether categorical variables appear related.
Class imbalance	One target class is much more common than another.
Confusion matrix	Counts true positives, true negatives, false positives, and false negatives.
Cramer's V	Association strength between categorical variables, from 0 to 1.
Cross-validation	Repeated training-data splits used to estimate model performance.
Decision tree	A model that predicts through if-then split rules.
EDA	Exploratory Data Analysis, used to understand data before modeling.
F1-score	A metric balancing precision and recall.
Feature	An input variable used by the model.
Feature importance	A model-specific score showing which inputs contributed most.
Hold-out test set	Data kept aside for final performance reporting.
IQR	Interquartile range, Q3 minus Q1.
KNN	K-nearest neighbors, a distance-based classifier.
Leakage	When answer-like information is accidentally included as input.
Macro-F1	F1 averaged across classes equally.
Naive Bayes	A simple probability-based classifier.
Neural network	A flexible model that learns patterns through connected layers.
One-hot encoding	Turning category labels into 0/1 columns.
Precision	Of predicted positives, the share that were truly positive.
Recall	Of true positives, the share the model found.
Stratified split	A split that preserves class proportions.
Target	The outcome variable the model predicts.
Target proxy	A variable that indirectly gives away the target.
Weighted-F1	F1 averaged across classes while giving larger classes more weight.